Prop. Let X be a Compact Hausdorff space. Then
$A \subseteq X$ is Compact if and only if $A \subseteq X$ is Closed

T_2

Compact.

Proof.

Prop

Let X be any space and Y be Compact Hausdorff space.

T_2

$f: X \to Y$ is Continuous map if and only if the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

Compact

proof. Let $f: X \to Y$ be a Continuous map.

Let $(x,y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets $\mathcal{U}, V$

$$\mathcal{U} \cap V = \phi, \quad y \in \mathcal{U}, \quad f(x) \in V$$

Since $f(x) \in V \overset{\text{def}}{\iff} x \in f^{-1}[V]$ and f is Continuous, $f^{-1}[V]$ is an open set in X.

Now, $(x,y) \in f^{-1}[V] \times \mathcal{U}$

And, $(f^{-1}[V] \times \mathcal{U}) \cap G_f = \phi$, because if $(x, f(x)) \in f^{-1}[V] \times \mathcal{U}$, then

$f(x) \in V$ and $f(x) \in \mathcal{U}$, it is contradiction to disjointness.

Therefore, $(x,y) \in f^{-1}[V] \times \mathcal{U} \subseteq X \times Y \setminus G_f$, so proved. $\rfloor \Rightarrow$

Conversely, let G_f is closed in $X \times Y$, under the Product Topology.

Let $C \subseteq Y$ be a Closed set in Y. Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $P_X : X \times Y \to X : (x,y) \mapsto x$ is closed map if Y is Compact. $\rfloor$ $\boxtimes$

$\Leftarrow$

Note: for $g = f \times i : X \times Y \to Y \times Y : (x,y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x,y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f$$

「Prop.

Compact Hausdorff space is Normal space.

proof. Let disjoint closed subsets C and D be given.

Let $x \in C$ be fixed. For each $y \in D$, since $C \cap D = \emptyset$, $y \neq x$.

By the T_2 condition, there exists disjoint open sets V_y and U_y, containing x and y respectively. Then, $\{U_y \mid y \in D\}$ is open cover for D and since D is closed set in the compact space, so it is compact. Therefore, there is finite subcover $\{U_{y_i} \mid i \in I\}$, where I is finite indices set. Put

$$V_x = \bigcap_{i \in I} V_y \quad \text{and} \quad W_x = \bigcup_{i \in I} U_y.$$

Since I is finite, V_x is open set containing x and and $V_y \cap U_y = \emptyset$ for all $y \in D$ implies $V_x \cap W_x = \emptyset$, and $D \subseteq W_x$.

Since $x \in X$ was arbitrary, we can construct an open cover for C,

$$\{V_x \mid x \in C\}$$

Similarly, C is compact, so there is finite open cover

$$\{V_{x_j} \mid j \in J\}$$

where J is finite. Now,

$$C \subseteq \bigcup_{j \in J} V_{x_j} \quad \text{and} \quad D \subseteq \bigcap_{j \in J} W_{x_j}$$ (It is also open since J finite)

and V_{x_j} is disjoint for some W_{x_k}, always thus these open sets are disjoint. ∎